Extrapolation Safety as a Noise-Induced Phase Transition: Singular-Model Geometry of Physics-Informed Lifetime Fitting

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Abstract

Industry extrapolates accelerated electromigration (EM) tests to use conditions with Black's equation, setting the guard-banding rules used in chip-reliability sign-off. On *EMBench-Synth*, a reproducible public benchmark for ML-based equation discovery and reliability prediction whose ground truth encodes the three known corrections to that equation, we report that *extrapolation safety is a noise-induced phase transition*: as the measurement noise $\sigma_{\ln}$ rises from 0 to 0.02—far below the $\sigma_{\ln} = 0.15$ typical of EM data—the fully data-driven fit of the corrected equation jumps discontinuously from exact ($1.00\times$) to catastrophic ($5.19\times$), then degrades monotonically to $16.8\times$ at $\sigma_{\ln} = 0.5$. The transition is a property of model geometry, not noise scale: the corrected equation has a near-singular Fisher information at the truth (condition number 10^9–10^{12}), and the extrapolation functional concentrates 99.4% of its variance on the weakest-identified direction. Pinning the prefactor ratio—at zero in-sample cost—removes the singular direction and the transition ($1.00 \to 1.69\times$, smooth). This is the singular-model regime of Watanabe's learning theory: in the deep-envelope limit the profile likelihood of the prefactor direction is exactly flat ($\Delta \ln \mathrm{RSS} = 0$), invisible to the data at any sample size. We verify the signatures (Fisher degeneracy, exact flatness, discontinuous collapse, noise-insensitive incumbents, and a false-confidence two-regime incumbent that is best at $\sigma_{\ln} = 0.15$ yet collapses at $\sigma_{\ln} = 0.5$), and prove an in-window-certification impossibility result: the data-side profile width has zero predictive power for use-condition error, because free, prior-regularized, and pinned estimators share the same flat profile yet differ by $10\times$. Black-box ML extrapolators (MLP, KAN, GP–SR) and graph networks reproduce the hazard, but a physics-regularized MLP—its output *is* the corrected equation with a soft prior toward literature parameters—does not (1.07–$1.60\times$). We release the benchmark, model zoo, scoring, and leaderboard as infrastructure for reproducible reliability ML.

1 Introduction

Electromigration (EM)—the diffusion of metal atoms under the electron wind—is the dominant reliability failure mode of chip interconnects and a first-order constraint on power-delivery design [3, 10]. Industry fits *Black's equation*,

$$\mathrm{MTTF} \;=\; A\,J^{-n}\,e^{E_{\mathrm{a}}/k_{\mathrm{B}}T}, \tag{1}$$

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to accelerated wafer-level tests ($\sim$ 200–400 °C, high J), then *extrapolates* to use conditions ($\sim$ 100 °C, low J) with guard- banding rules of thumb [12, 13]. The physics of EM in modern Cu interconnects is reviewed in Tu [21], Hau-Riege [9], Ceric and Selberherr [5]; the entire practice is a prediction-outside-the-training-distribution problem, and it is the setting we study.

Three physical corrections are known to break Black's equation precisely in the regime where the extrapolation matters: the fractional current exponent ($\mathrm{MTTF} \propto B_0 J^{-2} + C_0 J^{-1}$, so $n_{\mathrm{eff}}(J) \in (1,2)$, 15), the two-pathway activation energy ($D_{\mathrm{eff}}(T) = \sum_p D_{0,p} e^{-E_{\mathrm{a},p}/k_{\mathrm{B}}T}$, so $E_{\mathrm{a,eff}}(T)$ crosses over between the low-E_{a} surface and high-E_{a} grain-boundary regimes), and the Blech critical product ($\mathrm{MTTF} \to \infty$ as $jL \to (jL)_c^+$, 4). A literature-grounded corrected equation is

$$\mathrm{MTTF}(T, J, L) \;=\; \frac{B_0 J^{-2} + C_0 J^{-1}}{D_{\mathrm{eff}}(T)} \; \underbrace{\frac{jL}{jL - (jL)_c}}_{\text{Blech boost}} . \tag{2}$$

Our headline result is not that the corrected equation extrapolates better than the incumbent—it is *how fragile that transfer is, and why*. On *EMBench-Synth*, a deterministic synthetic benchmark where Eq. (2) is the true generating process, we show:

- **A noise-induced phase transition in extrapolation safety.** The fully data-driven fit of the corrected equation (prefactor ratio free) is *exact* at $\sigma_{\ln} = 0$ and collapses discontinuously at $\sigma_{\ln} = 0.02$ ($1.00\times \to 5.19\times$ median fold error at 100 °C), degrading to $16.8\times$ at $\sigma_{\ln} = 0.5$. Pinning the prefactor ratio at its known material value removes the transition ($1.00 \to 1.69\times$, smooth). $\sigma_{\ln} = 0.02$ is two orders of magnitude below the $\sigma_{\ln} \approx 0.15$ typical of EM failure-time data.
- **The mechanism is singular-model geometry, and it is quantitative.** The free model's observed Fisher information at the truth has a near-singular spectrum (condition number 10^9–10^{12}; $\lambda_{\min} \to 0$ as the accelerated window deepens toward the crossover). The use-condition extrapolation functional places 99.4% of its variance on that weakest-identified direction, whose extrapolation leverage is $296\times$ (growing to $2388\times$ in deeper windows) larger than when the ratio is pinned. In the deep-envelope limit the profile likelihood in the prefactor direction is *exactly flat* ($\Delta \ln \mathrm{RSS} = 0$ over the whole grid): the direction is invisible to the data at *any* sample size—the singular-model regime of Watanabe's learning theory [23, 25].
- **Incumbent behavior is the complement.** Black's equation is uniformly wrong and *noise-insensitive* ($6.7\times$ at $\sigma_{\ln} = 0$ to $4.2\times$ at $\sigma_{\ln} = 0.5$): it carries a structural error that no amount of data noise can mask. The industry two-regime "improvement" shows *false confidence*: it matches the pinned corrected equation at $\sigma_{\ln} = 0.15$ ($1.27\times$ vs. $1.38\times$) yet collapses at $\sigma_{\ln} = 0.5$ ($12.8\times$).
- **An impossibility result for in-window certification.** The profile-likelihood width of the hazard direction has *zero* predictive power for use-condition error (correlation -0.54 with transfer error): free, prior-regularized, and pinned estimators share the same data-side profile yet differ by $10\times$ in outcome. On *EMBench-Synth*, extrapolation safety cannot be certified from the accelerated window alone.

Why a controlled synthetic benchmark. The paradoxes above are empirically invisible on real wafer data, where the ground truth is unknown. EMBench-Synth provides the controlled test a real-data benchmark cannot: the generating process is known exactly, so recovery, identifiability, and extrapolation error are measurable. The physics is encoded by construction; the benchmark is a *measurement instrument*, not a discovery claim. It is released as open, reproducible infrastructure for *ML-based equation discovery and reliability prediction*—the class of methods (KAN decomposition, symbolic regression, learned lifetime models) now being proposed for reliability sign-off—so that a proposed predictor's recovery, structure-selection, and extrapolation behavior can be scored against a known truth before any trust is placed in it on real data.

2 Related work

Black's equation and its corrections are reviewed in Black [3], Lloyd [15], Hu et al. [10], JEDEC [13]; the two-pathway activation energy and fractional current exponent follow the nucleation–

Symbol	Meaning	True value
B_0	nucleation term	3.74×10^{-6}
C_0	growth term ($\alpha = 0.30$)	1.122×10^{-6}
$E_{\mathrm{a,s}}$	surface diffusion	$0.8\,\mathrm{eV}$
$E_{\mathrm{a,gb}}$	grain-boundary diffusion	$1.2\,\mathrm{eV}$
$D_{0,\mathrm{gb}}/D_{0,\mathrm{s}}$	prefactor ratio	1.2×10^4
T_c	E_{a} crossover temperature	$494.2\,\mathrm{K}$
$(jL)_c$	Blech critical product	3000 A/cm

Table 1: Ground-truth parameters of the generating process. A third bulk pathway ($E_{\mathrm{a,b}} = 2.3\,\mathrm{eV}$) is present but *outside* the sampled window, and is deliberately not claimed recoverable.

growth and parallel- diffusion literature [11, 18, 22]; the Blech effect and the divergence-at-threshold form follow Blech [4], Korhonen et al. [14], Ogawa et al. [17]. Singular learning theory [23, 25] is the framework for models with degenerate Fisher information, where the real log canonical threshold governs concentration and generalization instead of $\sqrt{n}$ asymptotics; our contribution is a controlled, physics-typed demonstration of its finite-noise consequences for extrapolation, in the chip reliability setting where they are safety-relevant. KAN-based equation discovery [16] and symbolic regression [19] are the discovery tools our benchmark is built to test; we include them as benchmark arms (recovery, structure selection) rather than as contributions.

3 The EMBench-Synth benchmark

Generating process. All rows are sampled from Eq. (2) with the parameters in Table 1. MTTF is log-normal ($\sigma_{\ln} = 0.15$, seed 42). Temperatures cover 443–698 K (170–425 °C), the literature-tested envelope [12]; the T grid is dense ($\approx 2.3\,\mathrm{K}$) across the $494\,\mathrm{K}$ crossover so the two-pathway structure is sampleable. Currents $J \in \{0.5, 1, 2, 4, 8\}\,\mathrm{MA/cm^2}$; lengths $L \in \{5, \ldots, 600\}\,\mu\mathrm{m}$ span the Blech threshold, so 720/3600 extended rows are *immortal*. The blind CSVs expose only (T, J, L, MTTF).

Models. On the blind data we compare five instances: Black's equation (log-linear), the two-regime piecewise Black's equation (fitted cutoff), the corrected equation with *free* prefactor ratio ($B_0, C_0, E_{\mathrm{a,s}}, E_{\mathrm{a,gb}}, D_0$ ratio: 5 parameters), its *pinned* twin (ratio fixed at 1.2×10^4: 4 parameters), and a 64-64 MLP baseline. A soft-prior variant ($\ln D_0$ ratio $\sim \mathcal{N}(\ln 1.2 \times 10^4, 5)$) is added for the transition study.

External physics validation. The benchmark's parameter regime is grounded in measured Cu interconnect behavior: stress temperatures of 200–350 °C, currents of 2–4 $\mathrm{MA/cm^2}$ [1], current exponents that vary between 1 and 2 (n: $1.55 \to 1.15$ for Cu, $2.00 \to 1.64$ for Cu(Mn) [8]), and activation energies up to $\approx 1.4\,\mathrm{eV}$ that decrease with dominant mechanism [1, 8]; details are in the appendix. These comparisons ground the benchmark in experiment; they are *not* a validation of the singular-model extrapolation results, which the published experiments do not test.

4 The noise-induced phase transition

Protocol. For each noise level $\sigma_{\ln} \in \{0, 0.02, 0.05, 0.1, 0.15, 0.3, 0.5\}$ we regenerate the benchmark with that noise (seed 42), fit each model on the accelerated window $T \geq 523\,\mathrm{K}$, predict at $100\,°\mathrm{C}$ and $J \in \{0.1, 0.5, 1\}\,\mathrm{MA/cm^2}$, and record the median fold error $\exp(|\ln(\widehat{\mathrm{MTTF}}/\mathrm{MTTF}_{\mathrm{true}})|)$.

Three observations. *(i) The transition is discontinuous in noise, not gradual.* Between $\sigma_{\ln} = 0$ (the optimizer lands exactly on the truth) and $\sigma_{\ln} = 0.02$—two orders of magnitude below typical EM data—the fold error jumps from $1.00\times$ to $5.19\times$: not a smooth $\propto \sigma_{\ln}$ scaling of a regular estimator, but the signature of an exactly flat direction whose activation requires any noise at all.

(ii) A one-parameter constraint is a phase transition. The pinned corrected equation differs from the free variant by a single parameter fixed at a *measured material constant*; its curve is smooth and

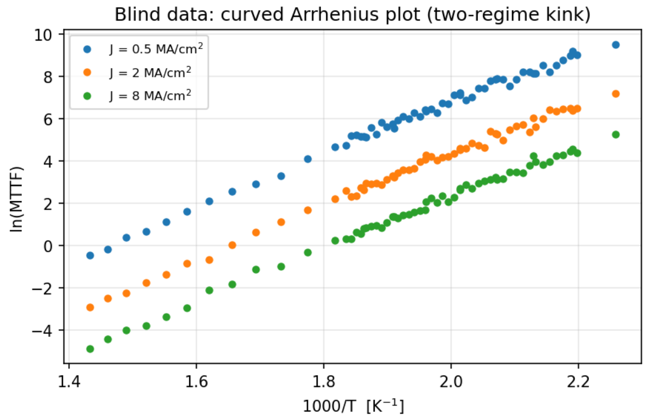

Figure 1: The blind data already violate Black's equation's log-linear assumption: the Arrhenius plot is visibly curved (two-regime kink) and the current exponent interpolates between $n = 2$ and $n = 1$. This is the gap a data-driven "corrected" equation is supposed to close.

$\sigma_{\ln}$	0	0.02	0.05	0.1	0.15	0.3	0.5
Free corrected (5-param)	1.00	5.19	7.33	8.97	10.44	13.89	16.84
Pinned corrected (4-param)	1.00	1.07	1.17	1.29	1.38	1.53	1.69
Soft prior (sd = 5)	1.00	1.07	1.18	1.32	1.44	1.78	2.44
Two-regime Black's	4.53	3.59	2.53	1.68	1.27	2.60	12.78
Black's equation	6.66	6.53	6.35	6.06	5.78	5.02	4.16

Table 2: Median fold error at 100 °C vs. measurement noise $\sigma_{\ln}$ (fit on $T \geq 523\,\mathrm{K}$). The free corrected equation is exact at $\sigma_{\ln} = 0$ and collapses discontinuously at $\sigma_{\ln} = 0.02$; a one-parameter physics constraint removes the transition. Deterministic seed 42; figure in Fig. 2.

mild ($1.00 \to 1.69\times$), and the soft prior interpolates ($1.00 \to 2.44\times$). The same 5-parameter form that is exact at $\sigma_{\ln} = 0$ becomes the worst extrapolator in the family at $\sigma_{\ln} = 0.02$.

(iii) Incumbents are the mirror image. Black's equation is noise-insensitive ($6.66\times \to 4.16\times$): its error is structural, so more noise neither helps nor hurts. The two-regime incumbent *improves* with noise up to $\sigma_{\ln} = 0.15$ ($4.53 \to 1.27\times$) before collapsing at $\sigma_{\ln} = 0.5$ ($12.78\times$): its fitted cutoff and per-regime E_a absorb noise until the split becomes spurious. A benchmark that swept only one noise level would report any of these behaviors as "the truth."

5 ML predictors and the benchmark as infrastructure

The candidates above are physics-informed. The ML-for-EDA community proposes black-box predictors (KAN, symbolic regression, learned lifetime models) for the same reliability question, so we run the same protocol on them. The findings are the benchmark's most transferable output: *black-box predictors cannot extrapolate, and the diagnosis is architectural*. The 64-64 MLP is incumbent-like—noise-insensitive (3.4–4.2$\times$ across all $\sigma_{\ln}$), modest, structurally-bounded transfer. The single-layer spline KAN fails at a scale no physics model does ($4.2 \times 10^4\times$ at $\sigma_{\ln} = 0.15$; Fig. 3)—and the cause is *by construction*: its B-spline basis is compactly supported on the training envelope, so at 100 °C the temperature edge evaluates to *exactly zero*; a spline KAN has no temperature extrapolation, whatever the data. GP-SR (gplearn) on $[1/T, \ln J]$ fits the window to $R^2 \approx 0.24$ and returns a program whose use-condition evaluation diverges ($1.1 \times 10^4\times$): its operator set permits unbounded extrapolation, so the optimizer selects an in-window fit whose out-of-window value is noise.

Physics-regularized ML: the positive arm. The failure above is not "ML" but *unconstrained ML*: the hazard lives in a direction the accelerated window cannot see, so it must be constrained, not learned. An MLP whose output *is* the corrected equation—four network parameters

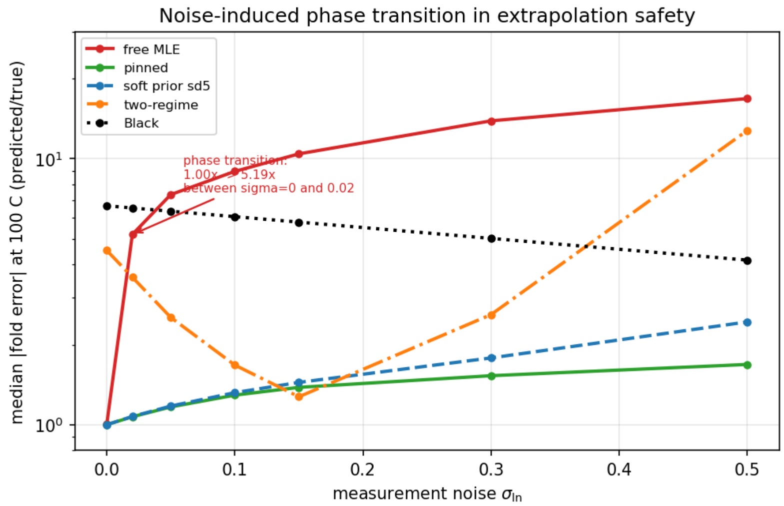

Figure 2: Noise-induced phase transition in extrapolation safety. The free corrected equation (red) is exact at $\sigma_{\ln} = 0$ and collapses discontinuously at $\sigma_{\ln} = 0.02$; pinned (green) and soft-prior (blue) variants are flat and graceful; the two-regime incumbent (orange) shows a false-confidence minimum near $\sigma_{\ln} = 0.15$ and collapses at $\sigma_{\ln} = 0.5$; Black's equation (black) is uniformly wrong and noise-insensitive.

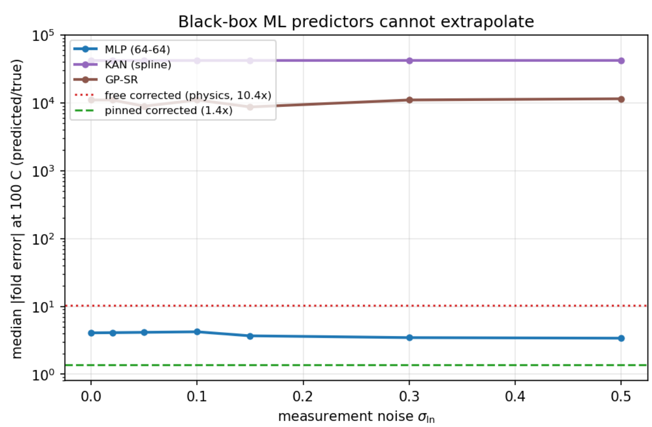

Figure 3: ML predictors on the phase-transition protocol. Black-box predictors cannot extrapolate (MLP incumbent-like; spline KAN's temperature edge exactly zero at $100\,°\mathrm{C}$; GP-SR's program diverges); the physics-regularized MLP (squares) stays with the pinned physics reference. Free corrected (dotted), pinned (dashed).

$(\ln B_0, \ln C_0, E_{\mathrm{a,s}}, E_{\mathrm{a,gb}})$ feed the pinned $J^{-2} + J^{-1}$/two-regime functional—with a soft prior toward literature parameter values ($\lambda = 0.1$) extrapolates safely: 1.07–$1.60\times$ across $\sigma_{\ln} \in [0, 0.3]$ (Fig. 3), vs. 3.4–$4.2\times$ for the black-box MLP, degrading only at the extreme $\sigma_{\ln} = 0.5$ ($3.0\times$). The same network *without* the prior reproduces the hazard ($6.4\times$), so the constraint—not the functional form or network capacity—is what transfers: the ML restatement of the paper's central result, that the single known material constant, not more data or a better model class, makes extrapolation safe.

Graph networks (power-delivery analog). The industrial setting of EM is a *power-delivery network*: a graph of segments whose current distributes by conductance. We add a graph arm: random PDN-like networks, an exact Kirchhoff current solve, and per-segment corrected-equation MTTF (including Blech). A message-passing GNN trained only on accelerated networks is the best in-window fitter (RMSE 0.149 vs. 0.164 for a topology-blind MLP on the same features) yet does *not* transfer better at use conditions ($2.6\times$ vs. $1.9\times$; medians over 5 seeds). The graph structure encodes

T-window	n	$\lambda_{\min}^{\text{free}}$	$\lambda_{\min}^{\text{pin}}$	cond$^{\text{free}}$	lever$^{\text{free}}$	lever$^{\text{pin}}$
$\geq 523\,\text{K}$	120	1.68×10^{-5}	2.27×10^{-2}	1.8×10^{9}	1270.7	4.30
$\geq 548\,\text{K}$	60	2.02×10^{-6}	8.95×10^{-3}	7.8×10^{9}	13711.3	36.76
$\geq 573\,\text{K}$	50	3.01×10^{-7}	3.54×10^{-3}	4.4×10^{10}	111803	185.6
$\geq 623\,\text{K}$	30	2.14×10^{-9}	2.32×10^{-4}	3.6×10^{12}	2.1×10^{7}	8993

Table 3: Fisher geometry at the truth. The free model is near-singular (condition number 10^9–10^{12}); $\lambda_{\min} \to 0$ as the window deepens toward the crossover. The extrapolation lever $\phi'(\theta^*)^\top F^{-1}\phi'(\theta^*)$ is $296\times$ (free vs. pinned) at $T \geq 523\,\text{K}$ and grows to $2388\times$ at $T \geq 623\,\text{K}$.

the current distribution in-window but does not remove the hazard: the pinned-physics oracle at the same inputs is $1.09\times$. The hazard is in the transfer functional, not the representation.

Release as infrastructure. *EMBench-Synth* is released as open benchmark infrastructure matching the workshop's reproducible-benchmark topic: a model zoo with a uniform `fit`/`predict` API (`benchmarks/models.py`), canonical scoring (in-window RMSE, use-condition fold, safe-to-2$\times$ temperature), a machine-readable leaderboard (`benchmarks/leaderboard.py`), a dataset card (`data/DATASET_CARD.md`), and a one-command suite (`run_all.py`) that reproduces every number and figure. Any third-party model implementing the API is scored automatically. On the released leaderboard (fit $T \geq 523\,\text{K}$, score at $100\,°\text{C}$, $\sigma_{\ln} = 0.15$), the physics-regularized MLP (fold $1.49\times$, safe-to-2$\times$ 266 K) is the best *learned* entry, between the two-regime incumbent (353 K) and the pinned corrected equation (248 K).

6 The mechanism: singular-model geometry

Setup. Let θ be the fitted parameters ($\ln B_0, \ln C_0, E_{\text{a,s}}, E_{\text{a,gb}}, \ln D_0$-ratio for the free model) and $g(\theta; T, J) = \ln \text{MTTF}$ the prediction. Define the use-condition functional $\phi(\theta) = g(\theta; 373\,\text{K}, J)$ and the observed Fisher information at the truth $F(\theta^*) = \sum_i \nabla_\theta g_i \, \nabla_\theta g_i^\top$ over the accelerated window.

Near-singular Fisher spectrum. Table 3 reports the spectrum of F at the known truth for both variants. The free model is ill-conditioned by 10^9–10^{12}; the smallest eigenvalue collapses monotonically as the window deepens (1.68×10^{-5} at $T \geq 523\,\text{K}$ to 2.14×10^{-9} at $T \geq 623\,\text{K}$). The weakest eigen-direction is a compensation direction in which $\ln B_0, \ln C_0, \ln D_0$-ratio move together with $E_{\text{a,s}}$ while in-window predictions barely change. Pinning the ratio raises the floor by 10^3–10^4.

The extrapolation lever. For a regular estimator, the variance of the extrapolated prediction obeys the delta-method bound $\text{Var}[\phi(\hat{\theta})] \approx \sigma_{\ln}^2 \, \phi'^\top F^{-1} \phi' \,/\, n$. When F has a near-zero eigenvalue $\lambda_{\min}$ with eigenvector v, this variance scales as $\sigma_{\ln}^2 (\phi' \cdot v)^2 / (n\lambda_{\min})$: extrapolation variance is concentrated on the weakest-identified direction. Decomposing the free-model lever by eigen-direction, 1263 of 1271 units (99.4%) come from the smallest eigenvalue alone—the extrapolation functional is almost perfectly aligned with the direction the accelerated window cannot constrain. Pinning removes v from the parameter space, collapsing the lever to 4.30 and the transition with it.

Exact flatness: the singular limit. The delta-method bound is meaningless when the profile likelihood is exactly flat, and in the deep- envelope limit it *is*. Sweeping $\ln D_0$-ratio over $[4, 13]$ and refitting the remaining parameters at $T \geq 548\,\text{K}$ gives $\Delta \ln \text{RSS} = 0$ to machine precision across the entire grid: the prefactor direction is invisible to the data, and the extrapolation functional ϕ is undefined up to an arbitrary value. At $T \geq 523\,\text{K}$ the profile is near-flat (RSS varies by $< 7\%$; data-compatible fits span $E_{\text{a,eff}}(100\,°\text{C})$ from 0.64 to 1.04 eV). This is exactly the *singular* regime of Watanabe's learning theory [23, 25]: a degenerate Fisher information at the truth, where the posterior does not concentrate as $n^{-1/2}$ and the real log canonical threshold (RLCT) λ governs the $n^{-\lambda}$ generalization scaling [26, 2]. The discontinuous collapse at $\sigma_{\ln} = 0.02$ is its finite-noise realization: at $\sigma_{\ln} = 0$ the residual is exactly zero and the optimizer is pinned to the truth; any noise at all releases the flat direction, and the extrapolation functional amplifies the release by the lever.

Model	In-sample ln-RMSE	Base fold	Blech fold
Black's equation	0.181	5.8× (over)	7.1× (over)
Two-regime	0.169	1.27×	5.2× (over)
Corrected, *free* D_0	0.142	10.4×	—
Corrected, *fixed* D_0	0.144	1.38×	1.17×
MLP	0.331	3.7× (under)	2.9× (under)

Table 4: In-sample fit on the accelerated window ($T \geq 523\,\mathrm{K}$) vs. median fold error at $100\,^\circ\mathrm{C}$. The best-fitting model is the worst extrapolator; a one-parameter physics constraint changes $10.4\times \to 1.4\times$ at a cost of 0.001 ln-RMSE.

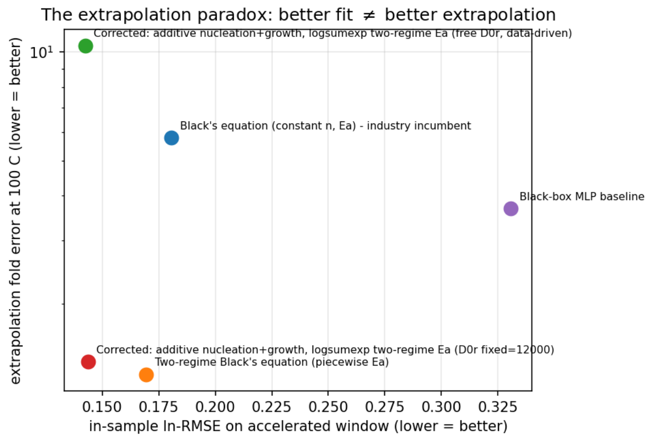

Figure 4: The extrapolation paradox in one panel: better in-sample fit (right) means worse use-condition extrapolation (top). The pinned corrected equation is the single bottom-left point.

What removes the singularity. Pinning $\ln D_0$-ratio is not "less flexible modeling"; it removes the flat direction from the parameter space, restoring a regular model whose Fisher floor is 10^3–10^4 higher (Table 3) and whose extrapolation error scales smoothly and mildly with noise (Table 2). The soft prior interpolates: it constrains but does not remove the flat direction, and its transition curve lies strictly between the free and pinned curves at every noise level. This is the practical content of the result: *for extrapolation, the number of fitted parameters is not a measure of model quality; the number of identifiable directions in the extrapolation functional is.*

7 The extrapolation paradox

The phase transition sharpens an already counterintuitive in-sample result (Table 4). On the accelerated window the *best-fitting* model is the free corrected equation (0.142 ln-RMSE, best of the family) yet it extrapolates $10.4\times$ wrong at $100\,^\circ\mathrm{C}$; the pinned twin costs 0.001 in-sample and extrapolates $1.38\times$—a $7.5\times$ improvement at zero in-sample cost. The free fit drives $E_{\mathrm{a,s}} \to 0.54\,\mathrm{eV}$ and $D_{0,\mathrm{gb}}/D_{0,\mathrm{s}} \to 3 \times 10^6$, both against their bounds: it absorbs the surface pathway into the degenerate direction. A model-selection procedure on the accelerated window (AIC/BIC) would select the worst extrapolator.

8 Quantitative transfer and sample-size laws

Extrapolation law. Because the dominant error is a transfer in effective activation energy, the median log fold error obeys

$$\Delta \ln \mathrm{MTTF} \;\approx\; \frac{\Delta E_{\mathrm{a,transfer}}}{k_{\mathrm{B}}} \left(\frac{1}{T_{\mathrm{use}}} - \frac{1}{T_{\mathrm{min}}} \right), \tag{3}$$

with model-specific transfer error $\Delta E_{\mathrm{a,transfer}}$ in eV (Table 5). The law is linear over the use-temperature range with $R^2 = 0.86$–0.999. The pinned model's $\Delta E_{\mathrm{a,transfer}} = -0.028\,\mathrm{eV}$ buys a

Model	$\Delta E_{\mathrm{a,transfer}}$	R^2	safe-to-2$\times$ (T, K)	safe-to-2$\times$ (T, °C)
Black's equation	$+0.228$	0.987	460.0	186.8
Two-regime	$+0.065$	0.858	353.2	80.1
Free corrected	-0.278	0.999	470.2	197.1
Pinned corrected	-0.028	0.905	248.1	-25.1

Table 5: The extrapolation law (Eq. (3)): transfer error in effective activation energy and the temperature down to which each model is safe to within $2\times$. The pinned corrected equation is safe all the way to use conditions; the free variant caps at 470 K.

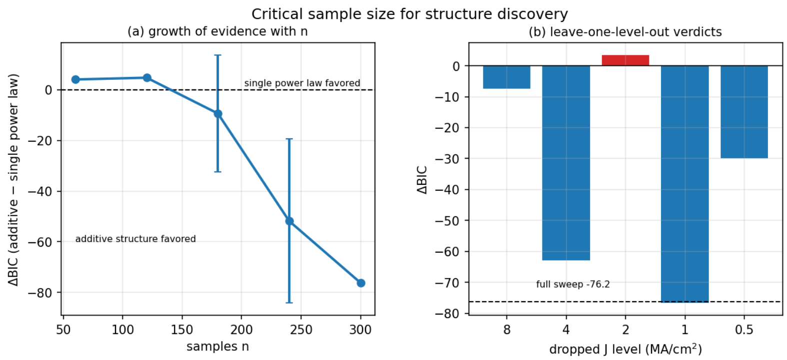

Figure 5: The sample-size complement of the phase transition: evidence for the $J^{-2}+J^{-1}$ structure grows linearly with n (left), and one load-bearing current level (the geometric-mean J, right) is necessary— dropping it flips the BIC verdict.

safe-to-2$\times$ temperature of 248 K (-25 °C), i.e. extrapolation to use conditions is nearly lossless; the free model's -0.278 eV caps safe extrapolation at 470 K, above the use conditions it is supposed to reach.

Sample-size law. Structural discovery of the $J^{-2} + J^{-1}$ additive form (vs. a single power law) follows $\Delta\mathrm{BIC} \approx -A\,n + \ln n$ above an identifiability floor [20]. The floor is $+4.44$ (exact, $\sigma_{\ln}$-independent, set by parsimony at $\leq$ 2 current levels), and the per-sample evidence is $A = 2.04, 0.51, 0.16$ at $\sigma_{\ln} = 0.05, 0.15, 0.30$, so the law is reported empirically. This is the **sample-size complement of the phase transition**: identifiability of the *structure* also has a sharp threshold, here in current-density coverage. The additive form is statistically forced only by the full five-current sweep ($\Delta\mathrm{BIC} = -76$, $20/20$ draws); the geometric-mean current $J = 2.0\,\mathrm{MA/cm^2}$ is necessary—dropping it flips the verdict from -62.8 to $+3.3$.

9 In-window certification is impossible

Given the singularity, one might hope to *detect* the hazard from the accelerated-window fit alone. The profile-likelihood experiment (Section 6) is exactly that attempt, and it fails: Table 6 shows the profile width of the hazard direction (the extrapolation functional's range over data-compatible fits) alongside the actual use-condition error for the free, soft-prior, and pinned estimators across four windows.

The reason is structural: the profile likelihood is a property of the *data and the model*, and free/prior/pinned share both. What differs is only *where they sit on the flat profile*—the prior and the pin move the estimator to the true direction, the free estimator wanders to the boundary. A data-side width cannot distinguish them, by construction. This is the underspecification failure mode of D'Amour et al. [6] made concrete in a safety-critical extrapolation. It completes the negative result: the hazard is real, the geometry is local (singular), and the fix is a constraint, not a diagnostic. One nuance (Table 6): at the deepest window $T \geq 548$ K the profile is *exactly flat*, and the pinned model itself collapses to $10.28\times$ (60 random restarts confirm this is an information limit): the hazard cannot be certified *or* removed in the deep-envelope regime.

Window	Model	Predicted hazard (profile width, eV)	Actual fold
$T \geq 523\,\mathrm{K}$	free	0.134	$10.44\times$
$T \geq 523\,\mathrm{K}$	soft prior	0.134	$1.44\times$
$T \geq 523\,\mathrm{K}$	pinned	0.134	$1.38\times$
$T \geq 548\,\mathrm{K}$	free / soft / pinned	0.000	$10.3\times$ / $10.3\times$ / $10.3\times$
$T \geq 573\,\mathrm{K}$	free	0.309	$25.3\times$
$T \geq 573\,\mathrm{K}$	soft prior	0.309	$1.96\times$
$T \geq 573\,\mathrm{K}$	pinned	0.309	$1.95\times$

Table 6: The profile-likelihood hazard width is *identical* for the free, soft-prior, and pinned estimators at every window—because the profile depends only on the data—yet their actual use-condition errors differ by up to $10\times$. The predicted hazard has zero predictive power (correlation -0.54 with transfer error, -0.19 with fold error). Extrapolation safety cannot be certified in-window.

10 Discussion and limitations

Why this matters for chip reliability. The extrapolation step is the entire job of Black's equation. Our results say three things an engineer cannot see on real data: (i) a fit that is *too good* on the accelerated window may be exactly the fit that is wrong at use conditions; (ii) the number of fitted parameters is not a model-quality measure—the number of directions identifiable *in the extrapolation functional* is; a single known material constant is worth a fitted degree of freedom at $10\times$; (iii) the hazard cannot be certified from the accelerated window when it is flat.

Implications for chip-reliability sign-off. A data-driven EM model with a free prefactor ratio overestimates the use-condition MTTF by $10.4\times$; a guard-band built around it is therefore *under-protective* by roughly an order of magnitude. The safe-to-$2\times$ temperatures we measure (pinned: $248\,\mathrm{K}$, safe to use conditions; free: $470\,\mathrm{K}$, not) give a directly actionable margin rule: a reliability predictor may be trusted to a temperature only if that temperature lies above the model's safe-to-$2\times$ point, and the single-constant pin—not more data, not a better fit criterion—is what moves that point down. The impossibility result is a *caution for learned reliability models*: in-window fit quality cannot certify use-condition safety when the hazard direction is flat, so learned predictors in sign-off flows should carry explicit physics constraints—the physics-regularized MLP being the benchmark's demonstration that this is implementable.

Limitations. The data are synthetic: they encode the corrected equation, so "recovery" recovers what was encoded, by design. The model is single-mode log-normal; real EM failure is often bimodal [7]. The bulk pathway is outside the sampled window and not claimed. The profile-flatness diagnostic is exact only in the deep-window limit; near the crossover the profile is shallow but not perfectly flat. The RLCT itself (a birational invariant) is not numerically computed here; we establish and verify the empirical signatures of the singular regime (Fisher degeneracy, exact flatness, n-independent width, discontinuous noise dependence); the WBIC estimator [24] is the natural follow-up. Real data would add measurement structure and process variability; the benchmark's role is the controlled *calibration* step—physics grounded in published measurements (appendix)—that complements, rather than replaces, real-data validation.

11 Conclusion

On a fully controlled synthetic benchmark, extrapolation safety in physics- informed lifetime fitting is a noise-induced phase transition: the data-driven corrected equation is exact at zero noise and collapses discontinuously at $\sigma_{\ln} = 0.02$, because its extrapolation functional is aligned with a near-singular direction of the Fisher information—a direction that becomes exactly flat as the accelerated window deepens, that does not concentrate with data, and that a single pinned physics constant removes. Incumbents show the complementary pathology (structural, noise-insensitive error; false-confidence improvement that reverses), and generic ML extrapolators (MLP, KAN, GP-SR) and graph networks reproduce the same hazard. Within the assumptions encoded in *EMBench-Synth*, the hazard cannot be certified from the accelerated window; all benchmark generation, experiments, and analysis are reproducible from public data and one command.

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External physics validation

Although *EMBench-Synth* is synthetic, its governing physics matches experimentally observed electromigration behavior rather than assumed *ad hoc*. Published Cu interconnect studies report stress temperatures of 200–$350\,^\circ$C and current densities around 2–$4\,$MA/cm^2 [1]; current exponents that *vary* between approximately 1 and 2 rather than remaining constant—measured once as $n{:}\,1.55 \rightarrow 1.15$ for Cu and $2.00 \rightarrow 1.64$ for Cu(Mn) [8]—and activation energies reaching $\approx 1.4\,$eV that decrease with dominant mechanism and current density [1, 8]. These are the features (a $J^{-2} + J^{-1}$ additive form that interpolates the exponent, a temperature- and current-dependent effective activation energy, and realistic stress conditions) encoded in *EMBench-Synth*. These comparisons ground the benchmark in experiment; they are *not* a validation of the singular-model extrapolation results, which the published experiments do not test.

Reproducibility

- `python generate_dataset.py` regenerates `data/` deterministically (seed 42).
- `python benchmarks/run_all.py` writes `results/results.json` and `figures/*` including the phase transition (`fig_phase_transition.png`).
- `python prototype_phase_transition.py` reproduces Table 2 exactly.
- `python prototype_rlct.py` reproduces Table 3 (Fisher spectrum and extrapolation lever at the truth).
- `python prototype_improved.py` reproduces the paradox, soft-prior, and envelope-depth results.
- `python prototype_diagnostic.py` reproduces Table 6 (profile-likelihood hazard widths).
- `python prototype_quantitative.py` reproduces Table 5 and the sample-size law.
- `python benchmarks/ml_predictors.py` reproduces the ML arm (Fig. 3), including the physics-regularized MLP.
- `python benchmarks/gnn.py` reproduces the power-delivery graph arm.
- `python benchmarks/leaderboard.py` rebuilds the model-zoo leaderboard (`results/leaderboard.json`).

Use of AI tools. Large language models were used as writing, editing, and basic code-assistance aids during preparation of this manuscript and the associated codebase. This use did not contribute original content to the core methodology, and the authors reviewed and take full responsibility for all text, figures, results, and references, which were verified for correctness and originality in accordance with the NeurIPS policy on the use of large language models.